DATA CHART GUIDE FOR FLIGHT

SCANFISH OPERATORS KEY TOOL

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Contents

1 Introduction 3

2 Data Chart introduction 3

3 Making adjustments based on observations in Data Chart 7

3.1 Too much cable out, low STW and/or shallow depth 8

3.2 Too little cable out, high STW and/or depth is too deep 9

4 Examples and actions 10

4.1 Balanced system 10

4.2 Flap locked out 11

4.3 Port and starboard flap separation 12

1 Introduction

This document is intended to give the ScanFish operator a guide of how to analyse the data from a **Data Chart** window in NaviSuite Flight and to make changes to the system that will give the ScanFish the best possible circumstances for being a stable platform and enhance manoeuvrability.

2 Data Chart introduction

The **Data Chart** window is opened within Flight via **View > Docable Panels > Data Chart**

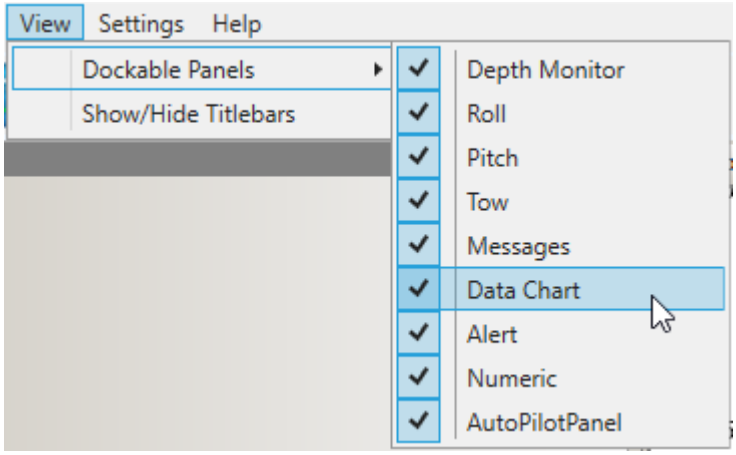

Figure 1 How to open a Data Chart window in Flight

The tick mark to the left of the **Data Chart** option in Figure 1 marks that one instance of the data chart is already open. It is possible to open multiple data chart windows, unlike many of the other windows in Flight that can only be opened once from this menu.

The window itself is a graph window where numerical values (y-axis) can be displayed as a function of time (x-axis).

The available numerical values that can be displayed are:

- **2D Flap Angle Port**: Displays the flap angles for port flap
- **2D Flap Angle Stbd**: Displays the flap angles for starboard flap
- **2D Flap Command Port**: Displays the command angle sent to the port flap
- **2D Flap Command Stbd**: Displays the command angle sent to the starboard flap
- **3D Flap Angle Port**: Displays the flap angles for port 3D flap
- **3D Flap Angle Stbd**: Displays the flap angles for starboard 3D flap
- **Altimeter**: Displays the altitude of the ScanFish
- **Battery** Voltage: Displays the ADC of the ScanFish
- **Cable Counter Speed**: Displays the speed in m/s of the winch cable
- **Cable Counter Value**: Displays the catenary length of winch cable
- **Controller Temperature**: Displays the hardware temperature in the controller
- **Distance Off Line**: Displays the cross-track error in metres
- **Heading**: Displays the ScanFish heading
- **Pitch**: Displays the ScanFish pitch
- **Roll**: Displays the ScanFish roll
- **Water Depth**: Displays the depth of the ScanFish

Combinations of these different numerical values can be arranged in the different open Data Chart windows.

Tip: A generally recommended combination to have open is the command and actual 2D flap motor angles, as shown in Figure 2 below:

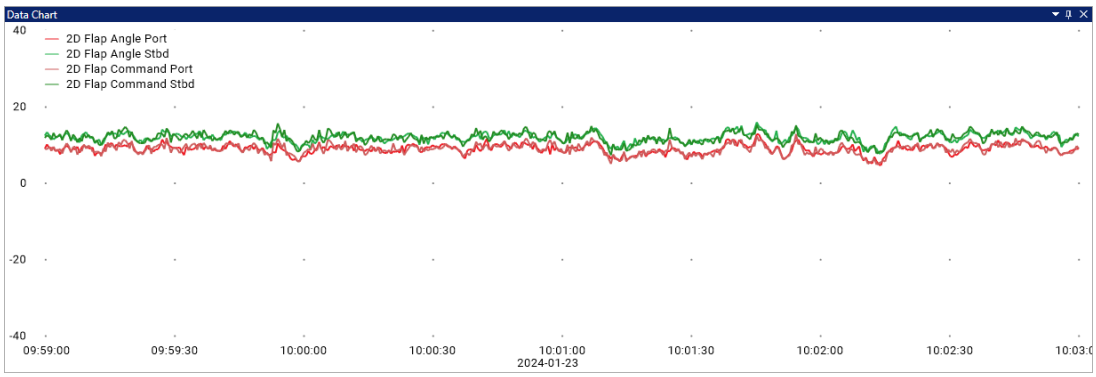

Figure 2 Data Chart combination of both flap motors' command and actual motor angles – Recommended to always have opened when operating the ScanFish

It is possible to make changes to the data chart window from **Settings > Option > User Interface > Data Chart Panel**. From here, the pen size and colour of each numerical value that can be displayed in a data chart can be changed.

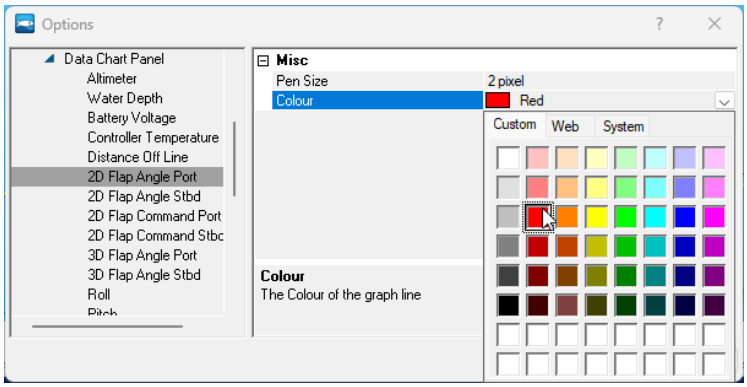

Figure 3 How to change graph colour in the Data Chart

Note: For the port and starboard flaps, it is recommended to have red and green colours respectively.

By selecting **Data Chart Panel,** the grid and background colours can be changed, together with axis settings (mode and min/max values) as well as legend location and the age of the data, as shown in Figure 4.

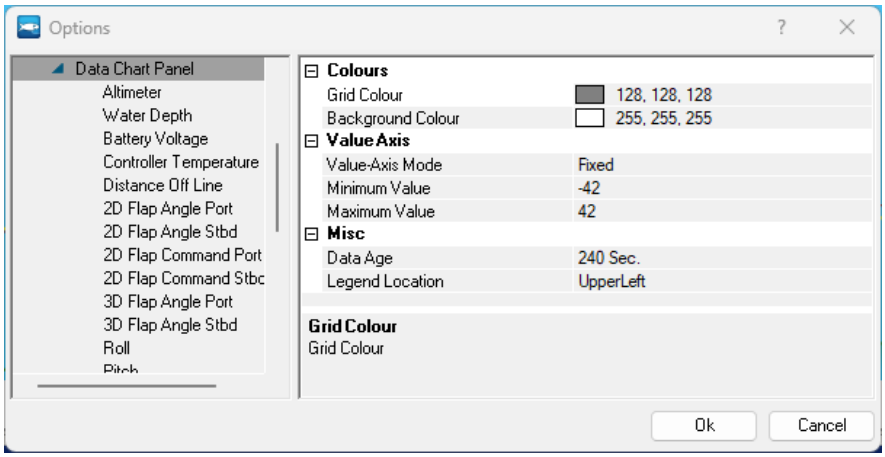

Figure 4 Settings for the Data Chart panel

The **Value-Axis Mode** has two options to choose from; **fixed** and **auto**.

It is recommended to have **Value-Axis Mode** set to **Fixed** mode for the flap motors and define values for **Minimum Value** set to *-42* and Maximum Value set to *42*.

The motors are limited to ±40 degrees during operation to prevent physical contact between the grub screw and end stop of the motor (read the ScanFish manual for more information on this at EIVA FreshDesk). By setting the **Minimum Value** and **Maximum Value** two degrees more (*-42* and *42*) means the user will be able to see when the motor is at its out most position in either of the two directions (as shown in Figure 5, where the port motor is locked out at constant 40° in the last ~45 seconds, where it becomes flat).

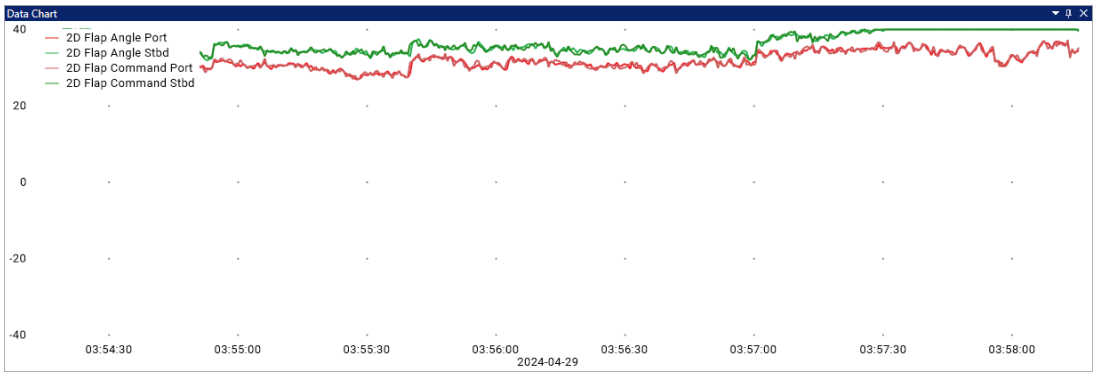

Figure 5 The starboard flap motor is at the maximum flap angle value

It is recommended to have the same data age in the **Data Chart** windown as in the **Depth Monitor** window, to easily compare flap changes with seabed changes between the two.

3 Making adjustments based on observations in Data Chart

A perfectly balanced system will have the flap angle curves flat and close to the neutral flap position (0˚) as shown in Figure 6.

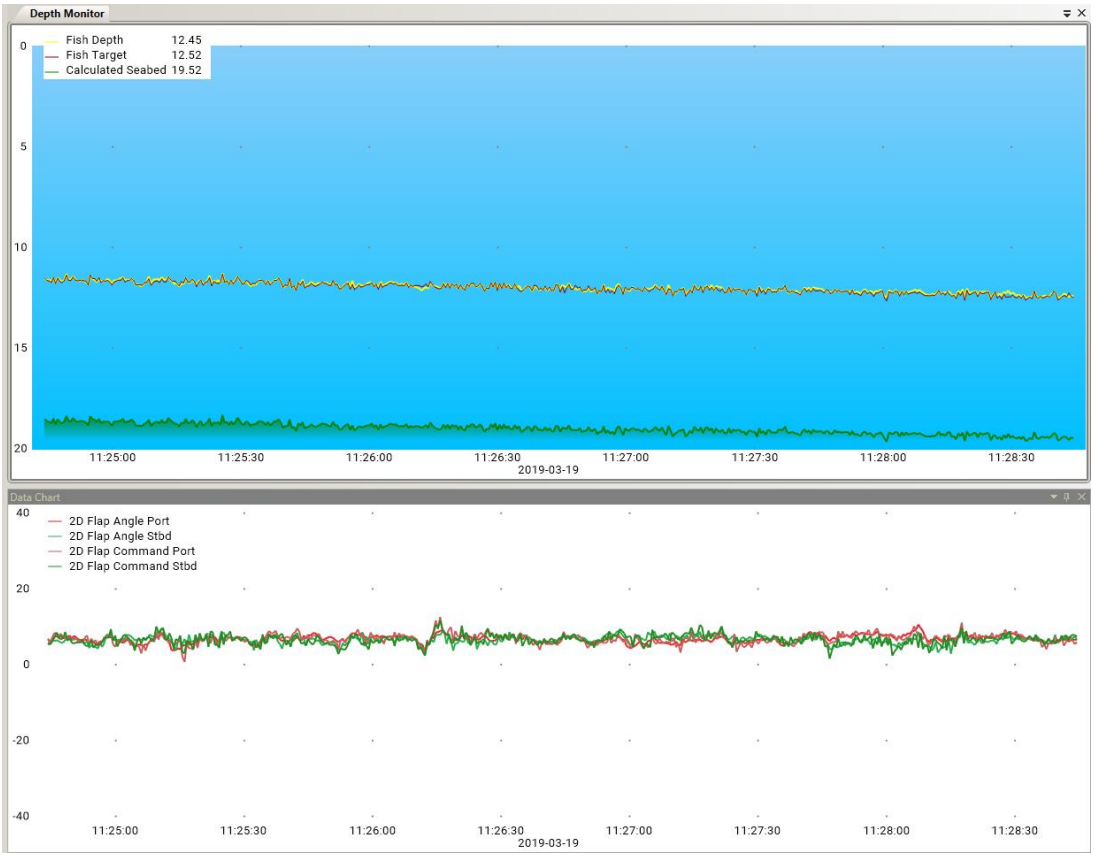

Figure 6 A well balanced system with flat flap angle curves around neutral position

In this scenario, the ScanFish needs to use minimal flap angle adjustments to stay at the desired depth, and only small flap adjustments for seabed topography changes. To have a perfectly balanced system the operator needs to find the right balance between the three factors that controls this, which are:

- Speed through water (STW)
- Cable length
- Survey depth

There are general guidelines in the ScanFish cable configuration document on FreshDesk, but it is up to the operator to be able to analyse the behaviour and adjust accordingly.

In many cases there will be a given survey depth for the area, therefore it is only the two remaining factors the operator can change for system optimisation. Some of the sensors installed with the ScanFish platform may also be limited to a certain speed over ground to get the data densities required. Consquently, the operator is limited to adjusting the cable length make changes to optimise the system performance – which also makes this a lot easier.

If the ScanFish is instructed to descend by the operator, or if seabed is changing to deeper depth, the motor flaps will respond by going to more positive angles than the starting point. If the ScanFish is instructed to ascend in the water column (either by the operator or shallower seabed), the flap motor angles will go to more negative angles than the starting point. This can be a bit counter intuitive, but it is important to remember:

- Positive values cause the ScanFish to descend
- Negative values cause the ScanFish to ascend

3.1 Too much cable out, low STW and/or shallow depth

If you have a ScanFish system that is tracking a relatively flat seabed, and the flaps have balanced themselves on flat curves in the negative section of the data chart (the yellow/red zone Figure 7), it should indicate to the operator that the ScanFish is constantly compensating for too much cable out, too low STW or shallow depth; it is actively trying to ascend with the flaps.

In that case, the operator should consider recovering some of the winch cable (at least if STW and survey depth are not changed) to move the curves from the yellow/red zone up to the green zone. In the green zone the flaps will be more efficient at adapting to seabed changes (both up and down) and give the system more manoeuvrability in general.

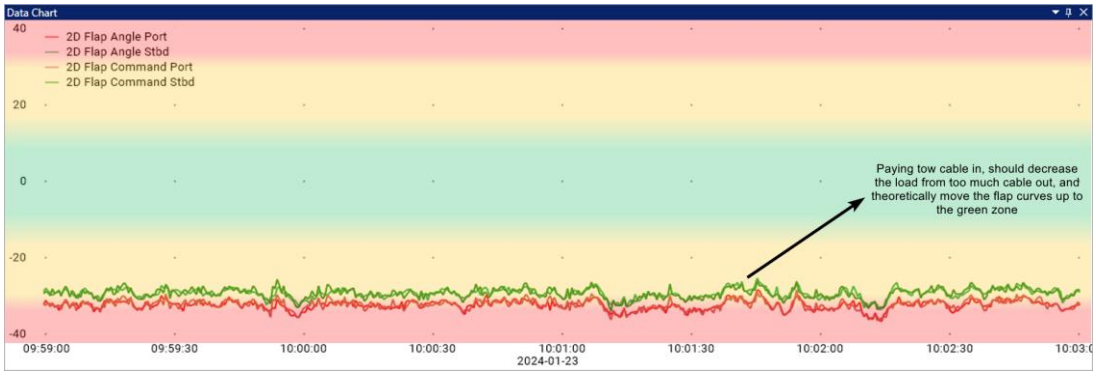

Figure 7 Data Chart with coloured zones for indicating good (green), medium (yellow) and bad (red) angle zones for the 2D flaps to be in

An alternative to paying cable in would be to increase the STW or go deeper with the ScanFish. That should both affect the system the same way as lowering the flap angles of the 2D motors.

3.2 Too little cable out, high STW and/or depth is too deep

If you encounter the situation where the flap curves have balanced themselves in the positive section as in Figure 8, the operator should consider either paying out additional winch cable, increase the STW or dive deeper to bring the flap angle curves from the yellow/red zone back to the green.

In the situation shown in Figure 8, there is also a difference between starboard (green) and port (red) flap curves, which could either be a mechanical (asymmetry in the system or somehow damaged/affected flap motor) or environmental (cross current) reason.

If it is mechanical, then the system should of course be recovered and fixed accordingly, but it is difficult to do anything from the operator's chair.

If it is due to cross currents introducing roll to the system, and the flaps try to compensate for it, then it should be the flap motor with highest angle (positive or negative), in this case the port side motor, that should be kept to a minimum.

Here, it would be recommended to pay additional cable out, which should give more positive flap angle to both motors, and ideally both should be inside the green zone.

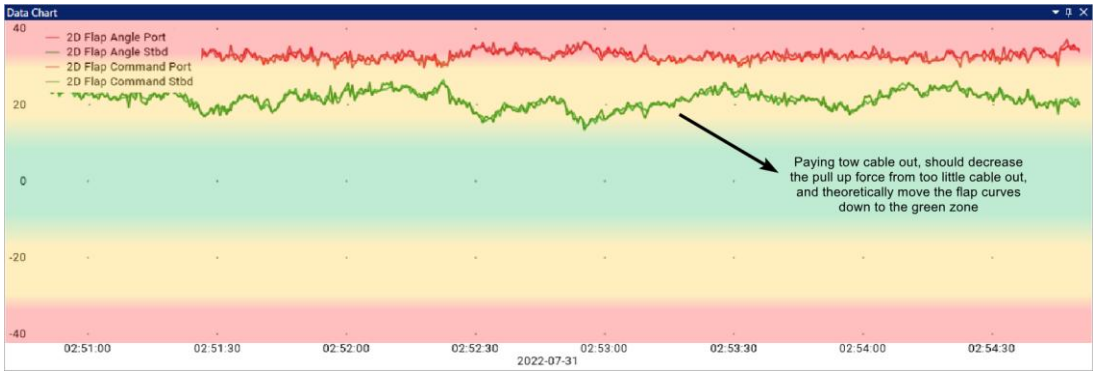

Figure 8 Data Chart with coloured zones for indicating good (green), medium (yellow) and bad (red) angle zones for the 2D flaps to be in

If working in a survey area where depths are constantly changing, the operator would need to make changes to the amount of winch cable payed out regularly to give the ScanFish the best possible conditions for optimal performance.

4 Examples and actions

4.1 Balanced system

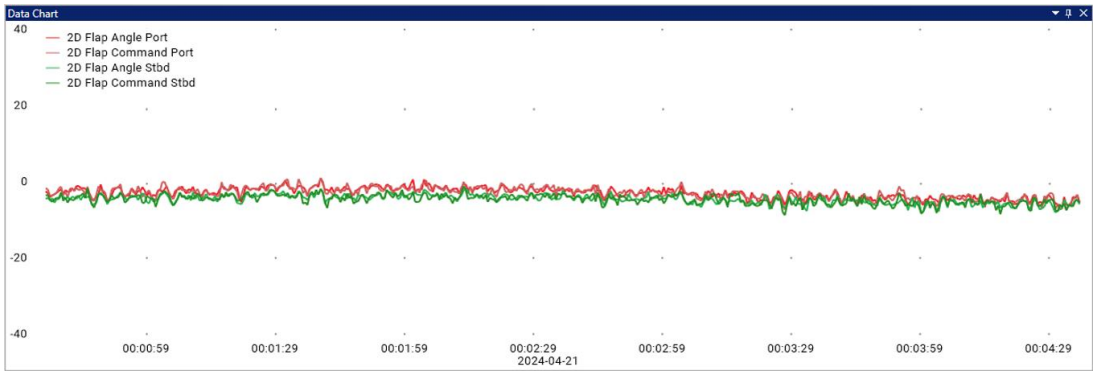

Figure 9 A well balanced system

An example of a well-balanced system. The curves are flat, only adjusting to seabed changes and both are close to 0˚. Here it can be assumed that there is a perfect relationship between depth, STW and cable length.

4.2 Flap locked out

Figure 10 Too-short cable out resulting in flap maxed out and roll issue

Figure 10 shows an example of a system that is not in balance.

This incident is from a follow seabed situation where the ScanFish tracks the seabed down to around 50 m depth, but due to the lack of cable to support the depth (140m out), the starboard flap maxes out, and as a result the ScanFish cannot keep itself stable anymore, and roll starts to build up and it is not following the target depth anymore.

At this point in time, the operator should quickly change the Flight mode to **Parking/Fixed depth** mode at a shallower depth than where the ScanFish is currently (eg. 30m) to allow the ScanFish to decrease some flap angle (ie the starboard flap comes back inside the operation window, and both flap curves gets closer to neutral position). That should bring the roll back to neutral as well. Before attempting to dive the ScanFish again, more cable should be payed out (if STW is as it should), allowing the ScanFish to go to the desired depth inside the flap angle operation window.

Note: In the example in Figure 10, the operator should have reacted to the big flap angle changes before the starboard flap angle got maxed out. These spikes indicate that something is not as it should, and the operator should have reconsidered cable length in relation to depth and speed earlier.

4.3 Port and starboard flap separation

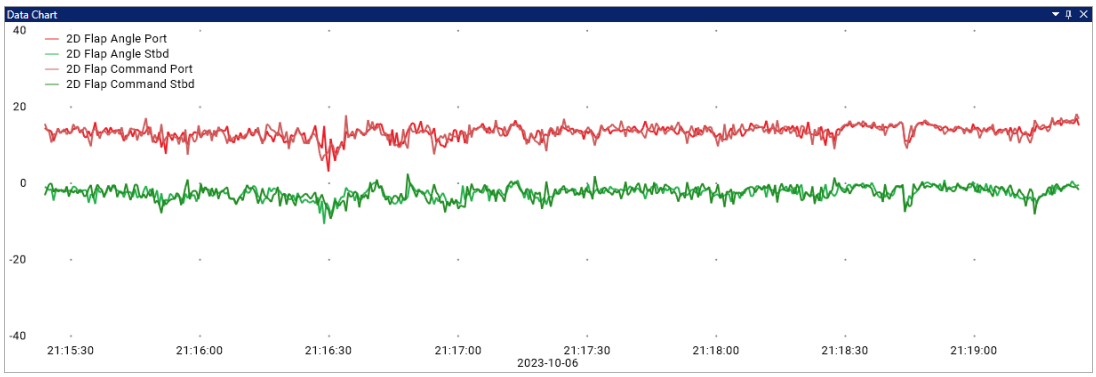

Figure 11 Side-current effect on an otherwise balanced system

Figure 11 shows an example of a reasonably balanced system.

The curves are slightly separated, which could be a result of a side-current that introduces some roll to the system, and the flaps are compensating for it because they are separated. If that is the case, and the separation is not due to anything mechanical, then it is hard for the operator to do anything about it.

Changing the cable length is likely the only thing that might solve this, as the starboard flap is more or less neutral, while the port flap is 15˚positive. By paying cable out, the flaps should both move to a slightly more negative average flap angle, and in this case would be perfectly positioned in -7.5˚ for the starboard flap, and +7.5˚ for the port flap – the 15˚separation is required to compensate for the roll that the side current introduces.